

How to utilise the limited supply of vaccines for Mpox control in Thailand among high-risk GBMSM

Sutham Jirapanakorn^a, Bannaporn Hunsin^b, Akira Endo^{d,e}, Clement Lee^f,
Shihui Jin^d, Borame Lee Dickens^d, Thitiya Theparod^{b,c,*}

^a Department of Disease Control, Ministry of Public Health, Nonthaburi, Thailand

^b Department of Mathematics, Mahasarakham University, Maha Sarakham, 44150, Thailand

^c Epidemic Simulation and Aetiology Nexus for Infectious Diseases (ESANID), Mahasarakham University, Thailand

^d Saw Swee Hock School of Public Health, National University of Singapore, Singapore

^e School of Tropical Medicine and Global Health, Nagasaki University, Nagasaki, Japan

^f School of Mathematics, Statistics and Physics, Newcastle University, United Kingdom

ARTICLE INFO

Article history:

Received 10 November 2025

Received in revised form 25 February 2026

Accepted 4 March 2026

Available online 4 March 2026

Handling editor: Dr Daihai He

Keywords:

Mpox control

Vaccination

Modelling

High-risk population

Post-exposure prophylaxis

ABSTRACT

Mpox re-emerged globally in 2022, disproportionately affecting gay, bisexual, and other men who have sex with men (GBMSM). In 2024, Thailand became the first Asian country to detect Clade Ib Mpox, prompting urgent decisions on deploying a limited supply of 3000 vaccine doses. However, evidence on the comparative effectiveness of different vaccine allocation and behavioural strategies in this context remains scarce. We developed a deterministic compartmental model of Mpox transmission among high- and low-risk GBMSM, calibrated to Thailand's national surveillance data (January 2023–May 2025). The model simulated a range of hypothetical scenarios under a constrained supply of 3000 vaccine doses, distributed either over a short 5-month period or extended across the 28-month epidemic horizon. We evaluated pre-exposure prophylaxis (PrEP), post-exposure prophylaxis (PEP), dose-sparing regimens, and mixed allocations of the two approaches. Each strategy was examined under alternative rollout timings (early vs. supply-delayed) and in combination with behaviour change, represented as reductions in sexual activity during symptomatic periods. The model reproduced Thailand's epidemic trajectory. Our simulations suggested that early PrEP rollout would have yielded the greatest reduction in incidence, particularly among high-risk GBMSM. PEP strategies would have had a modest impact overall, though single-dose sparing with delayed rollout (months 5–9) would have been notably effective as the epidemic peak occurred during this period. Mixed PrEP and PEP approaches would have produced intermediate benefits, while behaviour change alone significantly would have lowered transmission. Combining PEP with even modest behavioural changes further enhanced prevention and helped reduce spillover into low-risk groups. Under constrained vaccine supply, dose-sparing and mixed vaccination strategies could improve overall coverage and impact, especially when paired with behavioural changes. Integrating flexible and context-specific vaccination approaches with realistic behavioural modifications offers the best potential for Mpox control in Thailand and similar settings.

© 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

* Corresponding author. Department of Mathematics, Mahasarakham University, Maha Sarakham, 44150, Thailand.

E-mail address: thitiya.t@msu.ac.th (T. Theparod).

Peer review under the responsibility of KeAi Communications Co., Ltd.

1. Introduction

Mpox, a zoonotic disease caused by the monkeypox virus, has been historically endemic to Central and West Africa (Oliveira et al., 2017; Titanji et al., 2022). It is classified into two clades based on geographic origin: Clade I (Congo Basin) and Clade II (West Africa) (Alakunle et al., 2020; Srivastava et al., 2024). In 2022, a multi-country outbreak driven by a novel subclade of Clade II, classified as Clade IIb, disproportionately affected gay, bisexual, and other men who have sex with men (GBMSM), (Gostin et al., 2024; Okoli et al., 2024; Tiwari et al., 2025). In 2023, Clade Ib, another novel subclade originating from Clade I, emerged in the Democratic Republic of the Congo (DRC) and spread among wider population groups including sexually-active adults and children, raising more concerns for public health preparedness and response (Wenham & Eccleston-Turner, 2022). In October 2025, Clade Ib emerged in four European countries, prompting health agencies to call for renewed vigilance and targeted prevention measures (UK Health Security Agency, 2025).

The World Health Organization (WHO) declared Mpox as a Public Health Emergency of International Concern in July 2022 and again in August 2024 (World Health Organization, 2022, 2024). In November 2022, WHO issued interim guidelines recommending smallpox vaccines for Mpox prevention due to their cross-protective benefits (World Health Organization, 2024). Modified Vaccinia Ankara-Bavarian Nordic (MVA-BN) vaccines are recommended for use within these guidelines. The WHO also recommended prioritising *post-exposure prophylaxis* (PEP) for close contacts with Mpox cases and *pre-exposure prophylaxis* (PrEP) for high-risk populations, allowing countries to define high-risk populations based on their local epidemiological contexts (World Health Organization, 2022).

Thailand has been one of the particularly affected countries in Asia in the global mpox outbreaks with 888 cases as of May 16, 2025. Nearly 80% of clade IIb cases occurred among those self-identified as GBMSM individuals, with transmission mainly through sexual contact (Brooks et al., 2022). High-risk behaviours associated with transmission included having multiple sexual partners and engaging in sexual encounters with strangers (Jirapanakorn et al., 2024). In August 2024, Thailand became the first Asian country, and the second outside Africa, to report Clade Ib Mpox (BBC News, 2024). In response, the Department of Disease Control procured 3000 doses of MVA-BN with a budget of 21 million baht (USD 640,000). Thailand's vaccination strategy prioritised two main groups. The first group involve PrEP targeting individuals at occupational risk, such as medical personnel and laboratory staff involved in studying or treating Mpox. The second group involved individuals requiring PEP, such as those who had close contact with confirmed Mpox cases (Wipatayotin, 2024). Unlike the United States, where PrEP targeted at GBMSM (Deputy et al., 2023), Thailand's vaccination strategy prioritised PrEP for high-risk groups, while PEP is designated for individuals exposed to the virus (Frey et al., 2015; Mrsny, 2022; Pang et al., 2024; Shen et al., 2024). In this context, the exploration of dose-sparing strategies, according to a report of non-inferiority with only one-fifth of the original dose for MVA-BN (Dalton et al., 2023; Deputy et al., 2023; Frey et al., 2015; Kottkamp et al., 2023; Mrsny, 2022), might be a critical consideration for Thailand's limited MVA-BN stockpile.

Mathematical modelling provides a valuable framework for assessing epidemic dynamics and evaluating intervention strategies under such constraints. This study develops a deterministic compartmental model of Mpox transmission among high- and low-risk GBMSM, calibrated to national surveillance data. We evaluate the impact of alternative vaccination strategies, including PEP, PrEP, and dose-sparing, alongside behavioural changes, both individually and in combination, to inform evidence-based policy for Thailand's Mpox response.

2. Material and methods

We developed a deterministic compartmental model tailored to the GBMSM population. The model incorporates stratification by behavioural risk, natural history of infection, and two vaccination strategies: PrEP and PEP. We calibrated the model using national surveillance data and explored the influence of parameters through sensitivity analyses. Additionally, we simulated hypothetical intervention scenarios to assess the effectiveness of different vaccination strategies under conditions of limited vaccine availability, as well as the impact of behavioural changes in reducing transmission.

2.1. The transmission model

The model specifically characterises Mpox transmission among GBMSM in Thailand via sexual or intimate contact. A schematic representation of the model structure is presented in Fig. 1, and the definitions of model parameters are provided in Table 1. In the model, GBMSM is divided into high-risk and low-risk subpopulations based on the frequency of sexual activity. We assume that the average number of sexual contacts per month is 5 for the high-risk group and one for the low-risk group. The high-risk group comprises of individuals aged 20 years and older who are considered at an elevated risk of Mpox infection as they are sexually active (Department of disease control, 2025). Fig. 1 illustrates the model structure for the high-risk group only; the low-risk group follows the same disease progression framework but does not include vaccination compartments (PrEP and PEP). Transmission both within and between groups is modelled through group-specific contact rates, allowing for cross-group infections. Formally, the population is stratified into two distinct groups: high-risk individuals (denoted by h) and low-risk individuals (denoted by l). Disease transmission is governed by the force of infection (λ), which represents the per capita rate at which susceptible individuals acquire infection through contact with infectious individuals. It is defined as:

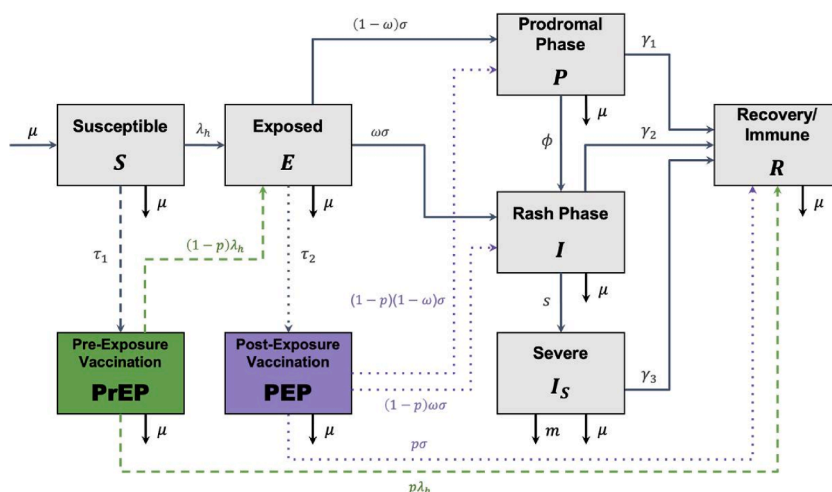


Fig. 1. Schematic diagram of the compartmental model used to simulate Mpox transmission dynamics and the impact of vaccination strategies. The population is divided into compartments: susceptible (S), exposed (E), prodromal phase (P), rash phase (I), severe symptoms (I_s), recovered/immune (R), and vaccinated groups through pre-exposure ($PrEP$) and post-exposure (PEP) prophylaxis. Transmission occurs via contact with infectious individuals in either the prodromal or rash phase. Vaccination is modelled through two pathways: $PrEP$ vaccination (at rate τ_1) and PEP vaccination (at rate τ_2). Individuals may develop symptoms after incubation, progressing through either a prodromal or direct rash phase depending on the probability ω . The model also incorporates progression rates ($\sigma, \lambda, \gamma_1, \gamma_2, \gamma_3$), severity (s), and death (m) with natural recruitment and natural death represented by μ .

$$\lambda_h = \beta_{hh}(P_h + \theta I_h) / N_h + \beta_{lh}(P_l + \theta I_l) / N_h,$$

$$\lambda_l = \beta_{ll}(P_l + \theta I_l) / N_l + \beta_{hl}(P_h + \theta I_h) / N_l,$$

where λ_h and λ_l are the forces of infection for the high-risk and low-risk groups, respectively.

β_{ij} ($i, j \in \{h, l\}$) denotes the effective transmission rate from infectious individuals in group i to susceptibles in group j . P and I refers to individuals in the prodromal phase and rash phase, respectively. θ is the proportion of symptomatic individuals (rash phase) who continue to engage in sexual activity, thus contributing to onward transmission. The full derivation and corresponding system of differential equations are provided in **Supplementary Material A**.

2.2. The course of infection

After infection, individuals are initially in an *exposed stage* (E), corresponding to the incubation period during which they are infected but not yet infectious. Following an incubation period, they may progress to an infectious state *prodromal phase* (P), characterised by the onset of mild, non-specific symptoms (e.g., fever, fatigue) that may not deter social or sexual activity, and subsequently to the *rash phase* (I). In this model, individuals are assumed to become infectious only if they are symptomatic, meaning that transmission begins during the prodromal phase and continues through the rash phase. A proportion of individuals (denoted by $1 - \omega$) first enter P . The remaining proportion (ω) bypasses the prodromal phase and enters I directly, which marks the onset of more apparent clinical symptoms and continued infectiousness. Individuals in P transition either to I or recover (R) without developing further symptoms. Those entering the rash phase may either recover or progress to a severe symptomatic phase (I_s), where some may die due to disease-related complications. In the severe stage, individuals are assumed to be non-sexually active and thus do not contribute to transmission. However, we assume that symptomatic but not severe individuals can still engage in sexual activity with a proportion of θ . Recovery can occur from any symptomatic stage; prodromal, rash, or severe, with different rates of transition. Recovered individuals move to the immune class (R), where they are assumed to have long-term immunity.

2.3. Vaccination

Vaccination is incorporated into the model via two pathways: $PrEP$ administered to susceptible individuals in the high-risk group, and PEP administered shortly after exposure but before symptom onset. Breakthrough infections may occur among vaccinated individuals for whom the vaccine fails, as we assume an all-or-nothing mechanism where only a proportion of vaccinated individuals are fully protected and transition directly to the recovered/immune class (R), without becoming infectious or contributing to further transmission. We assumed that vaccines are administered exclusively to individuals in the high-risk group, with a limited supply of 3000 doses.

Table 1
Model parameters used in the Mpox transmission model.

Parameters	Description	Value (month ⁻¹)	Reference
β_{hh}	Transmission rate between high-risk individuals	3.70	Fitted ^a
β_{hl}	Transmission rate from high-risk to low-risk individuals	0.17	Fitted
β_{lh}	Transmission rate from low-risk to high-risk individuals	$\beta_{lh} = \beta_{hl}$	Fixed
β_{ll}	Transmission rate from low-risk to low-risk individuals	$\beta_{hh}/2$	Fixed
μ	Per capita monthly rate at which individuals enter the sexually active adult population (sexual debut at age 20)	$\frac{1}{12 \times 58}$	Life expectancy at 20 years in Thailand: 58 (macrotrends, n.d.)
σ	Incubation period	$(1/5) \times 30$	(Centers for Disease Control and Prevention, 2024a; World Health Organization, 2024; Government of Canada, 2025; Government of Western Australia Department of Health, 2025)
ω	Proportion of infected individuals who do not experience the prodromal (pre-symptomatic) phase.	0.3	(Kröger et al., 2023; Lin et al., 2024; Reda et al., 2023)
ϕ	Progression rate from prodromal to rash phase	7.43	Fitted (Centers for Disease Control and Prevention, 2024a; Government of Canada, 2025; Government of Western Australia Department of Health, 2025)
s	Proportion of infected individuals will develop severe symptoms.	0.06	Okoli et al. (2023)
γ_1	Rate of recovery from the prodromal phase to the recovered state (R)	3.44	Fitted (Centers for Disease Control and Prevention, 2024b, 2025; World Health Organization, 2024)
γ_2	Rate of recovery from the rash phase to the recovered state (R)	$(1/21) \times 30$	(Centers for Disease Control and Prevention, 2024b, 2025; World Health Organization, 2024)
γ_3	Rate of recovery from the severe phase (I_s) to the recovered state (R)	$(1/28) \times 30$	(Centers for Disease Control and Prevention, 2024b, 2025; World Health Organization, 2024)
u	Reporting rate	0.15	Assumed
m	Monthly mortality rate among severe cases	0.29	Calculated ^b
τ_1	Monthly rate of PrEP vaccination	Varied across scenarios	Assumed
τ_2	Monthly rate of PEP vaccination	Varied across scenarios	Assumed
p	Vaccine Effectiveness	Varied across scenarios	(Dalton et al., 2023; Deputy et al., 2023; Mason et al., 2024; Pang et al., 2024)
θ	Proportion of symptomatic (in the rash phase) individuals who continue to engage in sexual activity	Varied across scenarios	(Lee et al., 2018; Lin et al., 2024; Richards et al., 2008)

^a The parameter β_{hh} represents the effective transmission rate between high-risk individuals and is defined as the product of the per-contact transmission probability (β) and the average number of sexual contacts per month (c_h). Given $c_h = 5$, the fitted value of $\beta_{hh} = 3.70$ corresponds to a per-contact transmission probability of approximately 0.74.

^b Calculated as a constant monthly hazard rate based on observed data. Among 872 reported infections, 13 deaths occurred. Assuming 6% of cases progress to severe symptoms, this corresponds to a 25% probability of death in the severe phase. Given an average duration of 28 days in the severe stage, the monthly mortality rate was calculated as $m = -\ln(1 - 0.25) \approx 0.29$.

2.4. Model fitting and parameterisation

The model was calibrated using monthly reported Mpox case data from Thailand's national notifiable disease surveillance (Department of disease control, 2025). The calibration period spanned from January 2023 to May 2025, covering the observed progression of the epidemic. Calibration was performed by minimizing the discrepancy between the model-predicted and the observed monthly incidence using a least squares approach. Specifically, the objective function minimised was the root mean square error (RMSE) between the observed incidence y_i and the model-predicted incidence $\hat{y}_i$, defined as $RMSE = \sqrt{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / n}$, where n is the total number of observations. Key model parameters estimated through this calibration included the proportion of the high-risk population (H_r), transmission rates (β_{hh}, β_{hl}), the progression rate from prodromal to rash phase (ϕ), recovery rate from the prodromal phase (γ_1). Because the model was formulated with monthly time steps to match surveillance data, all time-dependent parameters were converted to monthly rates. In the baseline model, we assumed no vaccination. At month 10 (November 2023), after the epidemic peak had passed, the population was assumed to have reduced risky sexual behaviour in response to increasing awareness. This behavioural

change was incorporated into the transmission coefficients: cross-group transmission (β_{hl} , β_{lh}) was reduced by 10% (to $0.9\beta_{hl}$; $0.9\beta_{lh}$), while within-group transmission among low-risk individuals (β_{ll}) was reduced by 20% (to $0.8\beta_{ll}$). In addition, the proportion of symptomatic individuals who continued sexual activity (θ) was reduced by 10%. A complete list of model parameters is presented in Table 1.

2.5. Sensitivity analysis

To evaluate the relative influence of model parameters on the cumulative number of Mpox infections, we conducted a global sensitivity analysis using Partial Rank Correlation Coefficient (PRCC). PRCC is a robust and efficient method for evaluating the impact of individual parameters on the model output over a global parameter space. We selected 15 key epidemiological and behavioural parameters for the analysis. Latin Hypercube Sampling (LHS) method was applied to the input parameters, and a total of 1000 simulations were generated. The primary model outcome used in the sensitivity analysis was the cumulative number of Mpox infections. PRCC indices were calculated using rank-transformed model outputs and parameters with 500 bootstrap replicates to assess robustness and uncertainty. The monotonicity assumption underlying PRCC was verified, with the diagnostic plots shown in Supplementary Fig. S1.

2.6. Hypothetical scenarios

To assess the public health impact of limited vaccine supply, we simulated a range of intervention scenarios targeting high-risk GBMSM populations. The primary objective was to evaluate how 3000 available doses could be most effectively deployed, while also considering the role of behavioural adaptations. We first explored vaccination-only strategies, which included post-exposure prophylaxis (PEP) using either standard two-dose, single-dose, or dose-sparing regimens, as well as pre-exposure prophylaxis (PrEP) targeting susceptible individuals in the high-risk group. In addition, we considered mixed strategies in which doses were distributed between PrEP and PEP in different proportions. For each vaccination strategy, we varied the timing of rollout. Limited rollout scenarios reflected the constraint of 3000 doses administered over a 5-month period, either early in the outbreak (months 1–5), as a preparedness-oriented scenario, or later (months 5–9) as a scenario of vaccine supply constraints, which coincided with higher incidence. Extended rollout scenarios allowed vaccination to continue over the full course of the simulation (months 1–28 or 5–28). In parallel, we examined behavioural change scenarios, modelled as reductions in sexual activity during the rash phase. Reductions of 15%, 25%, 35%, and 45% were simulated to capture a spectrum of possible behavioural responses after rash onset. These scenarios reflect evidence from previous outbreaks, where awareness and risk perception were associated with reductions in sexual activity. Finally, we evaluated combined strategies that integrated vaccination with behavioural change. In particular, we focused on supply-delayed PEP under a limited supply of 3000 doses, administered between months 5 and 9 due to vaccine availability constraints scenario, paired with a modest 15% reduction in sexual activity during symptomatic periods. This design reflects the practical context of Thailand's epidemic response and is consistent with WHO guidance to prioritise PEP in situations of limited vaccine availability.

3. Results

3.1. Model fitting

The model was able to reproduce the observed trajectory of the mpox outbreak in Thailand between January 2023 and May 2025 (Fig. 2), capturing both the timing and magnitude of the peak in mid-2023. Both observed and fitted Mpox cases showed a rapid increase during the first half of 2023 and peak in August, followed by a steady decline and sustained low-

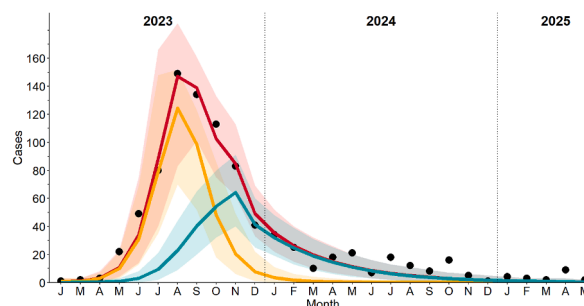


Fig. 2. Comparison between observed and model-simulated monthly Mpox cases in Thailand from January 2023 to May 2025. Black dots represent reported cases from national surveillance data. The solid red line shows the model-simulated total number of cases. The orange line represents cases originating from the high-risk group, while the teal line shows cases from the low-risk group. The model captures the peak and decline of the outbreak in 2023 and projects low-level transmission thereafter.

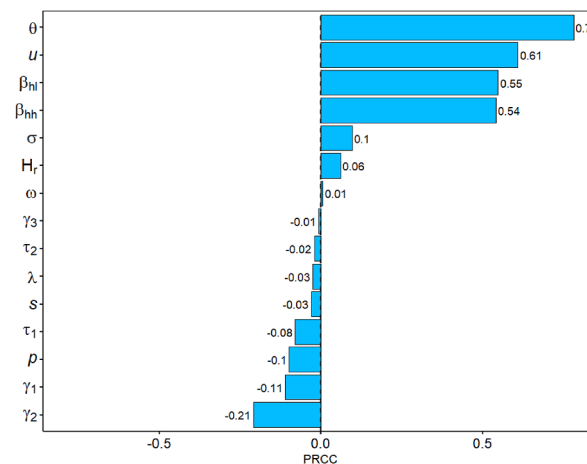


Fig. 3. Global sensitivity analysis using Partial Rank Correlation Coefficients (PRCC).

The bar plot shows the PRCC values for key model parameters with respect to the cumulative number of Mpox infections. Parameters with higher absolute PRCC values have a greater influence on the outcome. Positive values indicate that increasing the parameter increases the outcome, while negative values indicate an inverse relationship.

level transmission through early 2025. The high-risk group was estimated to account for 53.7% of total cases (95% confidence interval [CI]: 41.3%–60.2%), with the monthly incidence peaking in August 2023 at 124 cases (95% CI: 73–151). In contrast, transmission within the low-risk group was inferred to be of a smaller scale, where the estimated peak was delayed by approximately four months and 64 (95% CI: 41–93) cases were estimated to have occurred in that month. These modelled dynamics reflect both the behavioural risk stratification and the contribution of inter-group transmission in driving the outbreak.

3.2. Sensitivity analysis

Fig. 3 presents the results of the sensitivity analysis using Partial Rank Correlation Coefficients (PRCCs), showing the relative influence of model parameters on the cumulative number of Mpox infections. The analysis indicated that sexual activity during symptoms (θ), reporting rate (u), and transmission rates between groups (β_{hl}) and within the high-risk group

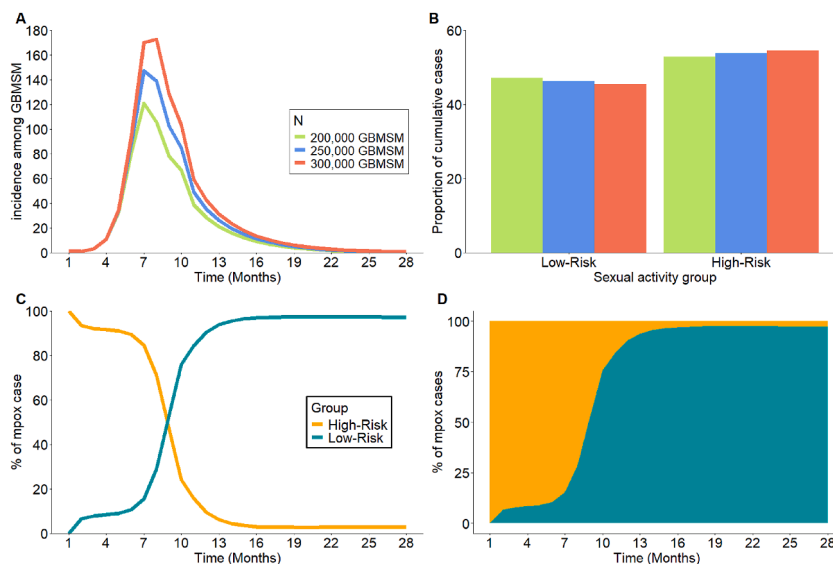


Fig. 4. Impact of GBMSM population size on epidemic dynamics. **A**, Incidence of Mpox among GBMSM, with difference population size, over time. **B**, Percentage distribution of cumulative Mpox cases between high- and low-risk groups. **C**, Temporal distribution of Mpox cases by risk group. **D**, Percentage of new Mpox infections by risk group. **A** and **B**, Results with GBMSM population sizes of 200,000, 250,000, or 300,000. **C** and **D**, Results with a GBMSM population of 250,000.

(β_{hh}) were the most influential parameters, each positively associated with epidemic size. In contrast, the recovery rates from the rash (γ_2) and prodromal (γ_1) phases exerted the strongest negative effects on Mpox infections.

Fig. 4 presents the sensitivity analysis of model outcomes to variations in GBMSM population size. The baseline scenario assumed 250,000 GBMSM, with alternative scenarios examining smaller (200,000) and larger (300,000) population sizes. Fig. 4A showed that although the overall scale of the epidemic increased with larger population size, the timing and shape of the epidemic curve remained consistent across scenarios, with incidence peaking at 7–8 months. Fig. 4B showed that the distribution of cumulative cases between high- and low-risk groups was relatively stable across population sizes, indicating that population size had little influence on the proportion of infections by risk group. Fig. 4C illustrated the temporal shift in the epidemic, with early cases concentrated in the high-risk group but, after around 8–10 months, the majority of infections occurring in the low-risk group. Finally, Fig. 4D highlighted the changing contribution of each group to new infections over time: high-risk individuals dominated transmission early in the outbreak, but low-risk individuals quickly became the main drivers of new cases as the epidemic matured.

3.3. Impact of vaccination

To further assess the potential impact of vaccination, we incorporated vaccination as an intervention applied throughout the entire simulation period. These scenarios were designed to evaluate how different strategies, namely pre-exposure prophylaxis (PrEP) and post-exposure prophylaxis (PEP), may influence epidemic dynamics. Fig. 5A–D shows the simulated impacts of varying monthly PrEP (τ_1) and PEP (τ_2) vaccination rates on Mpox incidence compared with the baseline scenario (no vaccination). Increasing PrEP coverage (Fig. 5A and C) was simulated to have substantially reduced incidence, and flattened and delayed the epidemic peak. At the highest PrEP rate ($\tau_1 = 0.1$), incidence would have been markedly reduced, with the peak delayed by several months. The violin plot (Fig. 5C) illustrates that higher PrEP coverage could have shifted the overall distribution of incidence downward, indicating sustained reductions in transmission across the entire simulation period. By contrast, increasing PEP coverage (Fig. 5B and D) may also have reduced incidence but with more modest effects. While higher values lowered the epidemic peak, the simulation timing of the peak was only slightly delayed compared with baseline. The violin plots (Fig. 5D) show that the distribution of incidence remained broader under the PEP scenario compared with PrEP, suggesting less consistent reductions in transmission. Taken together, these findings highlight that PrEP could be more effective than PEP in reducing transmission, delaying the epidemic peak, and lowering incidence consistently across the epidemic period. While PEP could provide some benefit, particularly at higher coverage, PrEP could yield a stronger and more sustained impact on epidemic control in high-risk GBMSM.

Building on these vaccination scenarios, we further examined how the timing of vaccine rollout influences epidemic dynamics when vaccination is restricted to a 5-month administration window (Supplementary Fig. S2). Fig. S2A and S2B show outcomes when vaccination was assumed to be introduced early, as a preparedness scenario, between months 1 and 5 of the epidemic. For PrEP, early vaccination was suggested to have been more effective because doses could have been given

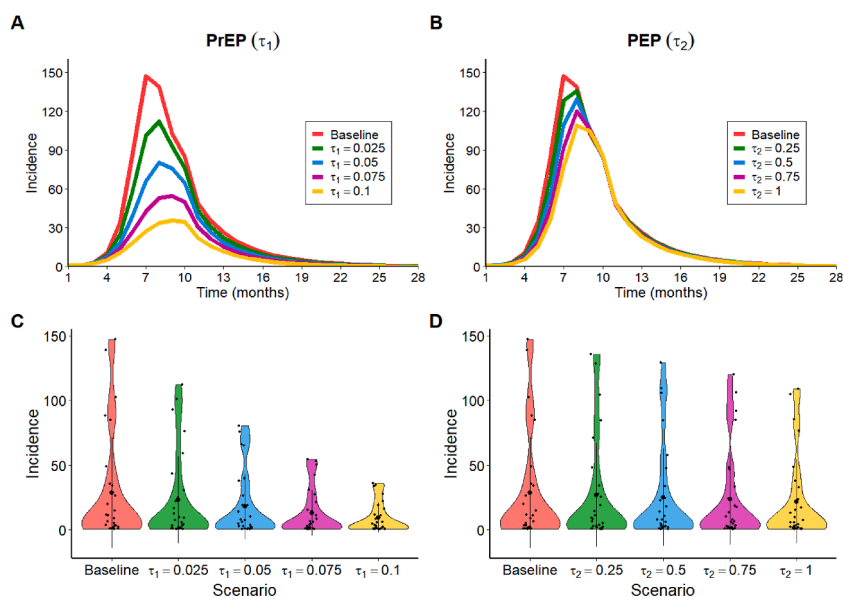


Fig. 5. Impact of pre-exposure prophylaxis (PrEP) and post-exposure prophylaxis (PEP) vaccination strategies on Mpox incidence. **A**, Incidence under varying monthly PrEP vaccination rates (τ_1). **B**, Incidence under varying monthly PEP vaccination rates (τ_2). **C**, Violin plots showing incidence distributions across simulations for different τ_1 values. **D**, Violin plots showing incidence distributions across simulations for different τ_2 values.

to susceptible individuals before they became infected, resulting in a stronger flattening and delay of the epidemic peak. In contrast, Fig. S2C and S2D illustrate the simulated impact of delaying vaccination until months 5–9. For PrEP, the delayed rollout would have had a weaker effect since much of the epidemic peak had already passed. However, for PEP, later implementation, reflecting vaccine availability constraints, would have coincided with higher incidence, meaning more exposed individuals could have been vaccinated at the right time, which enhanced its overall simulated impact. These findings suggest that timing requirements differ by vaccination approach: early rollout would be critical for PrEP, while PEP may achieve greater benefit when vaccine availability enables implementation during periods of higher transmission.

3.4. Impact of behaviour change

We next evaluated the potential impact of behavioural change during symptomatic periods and variations in reporting rates on Mpox transmission dynamics (Fig. 6). Reductions in sexual activity among symptomatic individuals (Fig. 6A and C) might have had a strong epidemiological effect. As the reduction factor (θ) increased, incidence declined sharply, the epidemic peak was substantially flattened, and overall transmission was suppressed.

At $\theta = 0.45$ (corresponding to a 45% reduction in sexual activity), incidence was almost eliminated, and simulated epidemic outcomes became consistently small across simulations, as illustrated by the violin plot distribution (Fig. 6C). By contrast, increasing the assumed reporting rate (u) primarily influenced the number of detected cases rather than the underlying epidemic dynamics (Fig. 6B and D). Higher reporting rates broadened the distribution of simulated incidence and increased the apparent epidemic size in surveillance data but did not directly alter transmission. These results suggest that while enhanced reporting is important for epidemic monitoring and case detection, reductions in risky behaviour during symptomatic periods could play a much stronger role in directly suppressing Mpox transmission.

3.5. Impact of limited supply of vaccination and behaviour change on high-risk and low-risk GBMSM

In this section, we further investigated the impact of vaccination under the constraint of a limited stockpile of 3000 doses, comparing outcomes in high-risk and low-risk GBMSM (Fig. 7). Four vaccination scenarios were evaluated: (i) standard two-dose, (ii) dose-sparing two-dose, (iii) single-dose, and (iv) dose-sparing single-dose. These strategies were implemented in our simulation under different rollout timings (early rollout: months 1–5 or across the full 28-month simulation; delayed rollout: months 5–9 or months 9–28) to assess how allocation might have affected prevention in each risk group.

Before running these scenarios, we explored the potential impact of dose-sparing vaccination under uncertainty in vaccine effectiveness (VE) and the proportion of a full dose administered (Fig. S3). Specifically, we assumed that fractional doses of 20%, 40%, 60%, and 80% of a full dose corresponded to VE values of 15.17%, 30.28%, 45.42%, and 60.56%, respectively. This sensitivity analysis showed that under PEP, varying VE and dose fraction had little effect on simulated epidemic outcomes. Based on these results, and to ensure consistency with prior work, we adopted VE values and dose-sparing

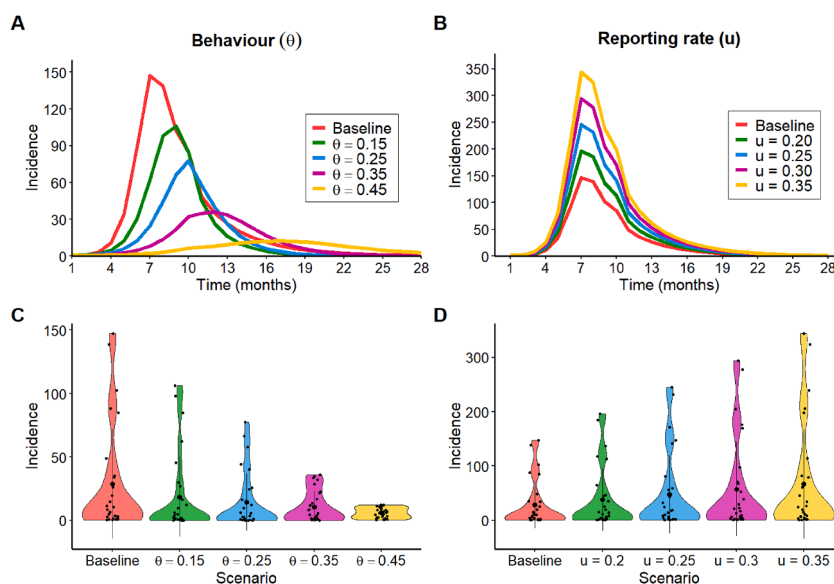


Fig. 6. Impact of behavioral change and reporting rate on Mpox incidence. **A**, Incidence under different levels of reduced sexual activity during symptomatic periods (θ). **B**, Incidence under varying reporting rates (u). **C**, Violin plots showing incidence distributions across simulations for different θ values. **D**, Violin plots showing incidence distributions across simulations for different u values.

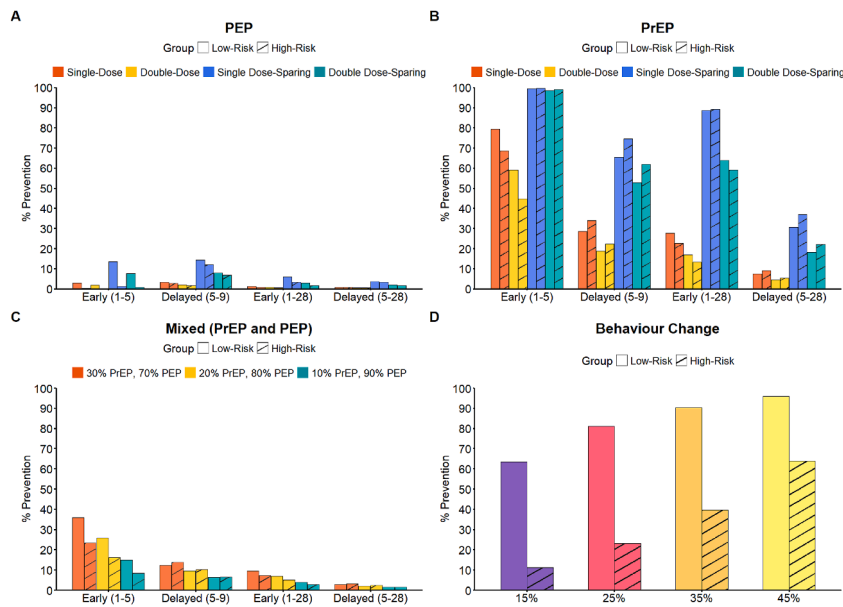


Fig. 7. Impact of vaccination and behavioral change under a limited supply of 3000 doses, with either short-term (months 1–5 or 5–9) or extended rollout (months 1–28 or 5–28). **A**, PEP strategies. **B**, PrEP strategies. **C**, Mixed PrEP and PEP strategies. **D**, Behavioural change alone. Scenarios are shown for both high- and low-risk GBMSM.

assumptions from the literature (Dalton et al., 2023) for the main vaccination scenarios presented in Fig. 7. A detailed summary of the scenarios and their parameters is provided in Supplementary Table S1.

Overall, PrEP strategies (Fig. 7B) achieved the greatest reductions in simulated incidence, particularly when implemented early (months 1–5). Across most rollout timings, the simulated impact of PrEP was similar between low-risk and high-risk groups, although delayed rollout (months 5–9) produced a greater reduction in the high-risk group. For PEP (Fig. 7A), the single-dose sparing strategy with delayed rollout (months 5–9) achieved the largest overall reduction in simulated total incidence, whereas early rollout (months 1–5) of single-dose sparing PEP provided disproportionately large benefits in the low-risk group. Mixed strategies combining PrEP and PEP (Fig. 7C) produced intermediate levels of prevention, with the 30% PrEP and 70% PEP allocation yielding the greatest overall impact. Finally, behavioural interventions alone (Fig. 7D) substantially reduced simulated incidence, with prevention increasing in proportion to the level of behavioural change, rising from 15% to 45% (detailed percentages of cases prevented with 95% CIs are provided in Supplementary Table S2). Together, these results highlight that under constrained vaccine supply, both vaccination strategies and behavioural change could significantly reduce transmission across high- and low-risk GBMSM populations.

3.6. Impact of combined PEP vaccination and behavioural change on high-risk and low-risk GBMSM

Given the WHO recommendation to prioritise PEP when vaccine supply is limited, we next examined the combined impact of PEP vaccination and behavioural change on epidemic outcomes in low-risk and high-risk GBMSM (Fig. 8 and

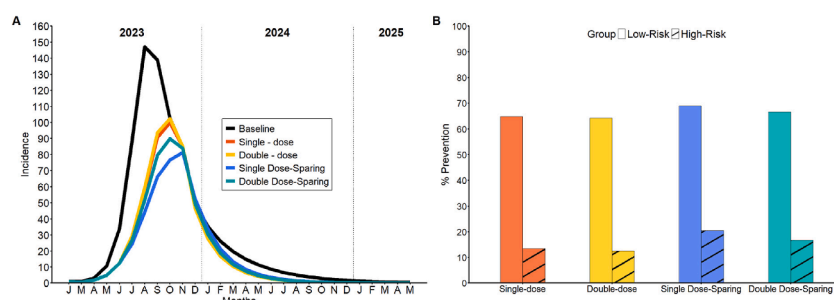


Fig. 8. Combined impact of PEP vaccination and behavioural change. **A**, Incidence trajectories under different PEP strategies with a 15% reduction in sexual activity during symptomatic periods. **B**, percentage of prevention by risk group under each strategy. Scenarios assume supply-delayed rollout (months 5–9) with a limited stockpile of 3000 doses in high- and low-risk GBMSM.

Supplementary Table S3). In this analysis, a 15% reduction in sexual activity during symptomatic periods was used as the baseline level of behavioural change and combined with different PEP strategies. For all PEP scenarios, we assumed vaccine supply-delayed initiation beginning 5 months after epidemic onset, with a limited supply of 3000 doses administered over a 5-month period (month 5–9). Across all regimens, combining PEP with behavioural change substantially enhanced prevention compared with either strategy alone. Each PEP strategy combined with 15% behavioural change delayed the simulated epidemic peak and lowered its magnitude (Fig. 8A), with the single-dose dose-sparing strategy achieving the greatest reduction. The bar plots (Fig. 8B) highlight clear differences across risk groups: low-risk GBMSM experienced the largest relative benefits, with prevention levels exceeding 60% across all strategies (detailed percentages with 95% CIs are provided in Supplementary Table S3). In contrast, reductions among high-risk GBMSM were more modest, remaining below 20.5%. This pattern reflects the higher contact rates assumed in the high-risk group, which limit the effectiveness of PEP alone. Taken together, these findings suggest that while PEP vaccination combined with behavioural change could be effective in curbing overall transmission, particularly by reducing spillover into the low-risk group, its direct protective effect for high-risk GBMSM may remain limited. This highlights the need to complement PEP with additional interventions, such as targeted PrEP or stronger behavioural modifications, to achieve meaningful reductions in incidence within the high-risk core group.

4. Discussion

In this study, we developed a mathematical model to characterise Mpox transmission dynamics and evaluated the possible impact of different vaccination strategies (PrEP and PEP), dose-sparing approaches, and behavioural change on outbreak outcomes among GBMSM in Thailand. We found that PrEP strategies might have achieved the greatest overall reductions in incidence, particularly when implemented early, reflecting the benefit of directly protecting susceptible high-risk individuals before infection and indirectly reducing transmission to low-risk groups. In contrast, PEP alone might have produced more modest reductions, as it relies on case identification, contact tracing, and vaccination after exposure. Nevertheless, the single-dose sparing PEP strategy with vaccine supply-delayed initiation (months 5–9) might have yielded the largest overall reduction in total incidence by coinciding with the epidemic peak when many exposed individuals were present. Interestingly, early rollout (months 1–5) of single-dose sparing PEP was simulation to provide disproportionately large benefits in the low-risk group, suggesting indirect protection when spillover from high-to low-risk GBMSM could have been prevented. Mixed strategies that combined PrEP and PEP were simulated to achieve intermediate levels of prevention, with the 30% PrEP and 70% PEP allocation producing the greatest overall benefit. Finally, behavioural change alone had a strong effect, with even modest reductions (15–25%) in sexual activity during symptomatic periods substantially lowering transmission. Combining vaccination with behavioural change could have amplified prevention further, with the largest simulated relative gains observed in the low-risk group. Notably, the estimated concentration of cases in the high-risk group (approximately 53.7%) emerged from model calibration rather than being imposed as a fixed modelling assumption. While uncertainty in this estimate remains - owing to surveillance limitations, reporting biases, and imperfect classification of behavioural risk - the qualitative conclusions of this study remain robust. Under limited vaccine supply constraints, prioritising individuals with higher contact rates consistently yields greater epidemiological impact per dose than untargeted allocation. If transmission were less concentrated within the high-risk group, the magnitude of benefit from highly targeted strategies would be reduced; however, the overall direction of policy recommendations would remain unchanged. These findings are consistent with prior studies (Granskog et al., 2025; Huang et al., 2024) and provide several important public health insights. First, incorporating PrEP for high-risk GBMSM into vaccine allocation, rather than limiting PrEP to occupational groups, could offer greater epidemic control compared to PEP alone. This supports WHO recommendations to prioritise specific high-risk populations, including GBMSM, for PrEP when feasible. However, given global vaccine shortages, current WHO guidelines advise prioritising PEP, with PrEP considered based on local epidemiological conditions (World Health Organization, 2022). However, in practice, some degree of vaccine leakage from the intended high-risk group to lower-risk individuals is possible due to stigma, imperfect risk disclosure, and programmatic constraints. Epidemiologically, such leakage leads to a predictable trade-off: while it can reduce infections in the low-risk group, it diminishes the effectiveness of PrEP in the high-risk core group that disproportionately contributes to onward transmission. To examine the robustness of our conclusions under more realistic implementation conditions, we conducted an additional sensitivity analysis allowing partial misallocation of vaccine doses (10–40%) to low-risk group. The results indicate that a small degree of leakage ($\approx 10\%$) can substantially increase protection in the low-risk group while maintaining comparable population-level epidemic reduction. However, higher levels of leakage produce diminishing returns once protection in the low-risk group approaches saturation. Beyond this point, additional misallocation primarily redistributes available vaccine doses rather than generating further epidemiological benefit, as fewer doses remain available for the high-risk group. These findings suggest that vaccination strategies are relatively robust to minor operational inefficiencies, but sustained prioritisation of high-risk groups remains important to maximise the overall effectiveness of Mpox vaccination programme (Supplementary Fig. S4). Our results also suggest that dose-sparing and single-dose PEP strategies can outperform standard two-dose regimens, despite lower individual-level effectiveness, by extending coverage to more individuals. This finding aligns with evidence from previous outbreaks (Dalton et al., 2023; Deputy et al., 2023; Dimitrov et al., 2023; Pang et al., 2024) and underscores the importance of optimising vaccine allocation under constrained supply. In Thailand, implementing dose-sparing via intradermal administration is operationally feasible, as this technique is already familiar to

clinical personnel through routine practice with other vaccines (e.g., BCG and rabies). Practical challenges may instead relate to communication and acceptability, including public skepticism or misinformation regarding the safety and effectiveness of an off-label administration route (Mrsny R. J., 2022), as well as local adverse event at the injection site that may raise cosmetic concerns or perceived stigma (Frey et al., 2015; Shen et al., 2024). These issues could be mitigated through pre-vaccination counselling, and strengthened monitoring and guidance for managing adverse events following immunization. Second, while vaccination remains central to outbreak control, it may not be sufficient by itself to curb transmission in the context of limited vaccine availability (Huang et al., 2024; Maniscalco et al., 2024). Our results suggest that behavioural interventions, particularly reductions in sexual activity during symptomatic periods, could substantially reduce incidence and, when combined with vaccination, further enhance epidemic control. This aligns with earlier work (Lin et al., 2024) and, highlights the critical role of behavioural change in mpox outbreak control.

From a policy perspective, these results provide useful guidance for Thailand and other countries facing similar constraints. Although our modelled results indicated that PrEP could offer the largest potential benefit, particularly under an early rollout (1–5 months), preparedness-oriented scenario in which vaccine is available soon after outbreak recognition, its feasibility is limited because vaccine access is typically reactive and may be delayed following case detection due to procurement lead times, importation logistics, and broader supply constraints during periods of global scarcity. Therefore, preparedness efforts such as establishing advance stockpiles and initiating procurement as early as possible following risk assessment of outbreak importation may be critical to enable timely PrEP implementation. The WHO currently recommends prioritising PEP in such situations, and our findings suggest that combining PEP with behavioural change can be an effective strategy. Notably, the high levels of prevention simulated in the low-risk group emphasise the population-wide benefits that could be achieved even when vaccines are targeted primarily at high-risk individuals. Encouraging reductions in sexual activity during symptomatic periods may therefore represent a highly cost-effective complement to vaccination, particularly in contexts where vaccine supply is limited or delayed.

Several limitations of this study should be acknowledged. First, the model necessarily simplifies Mpox transmission by stratifying GBMSM into only two risk groups and focusing exclusively on sexual or intimate contact. Other transmission pathways, such as household or occupational exposures, were not explicitly modelled. Second, estimates of vaccine effectiveness, particularly under dose-sparing regimens, remain uncertain. Although sensitivity analyses were conducted and informed by the best available evidence, these parameters may change as additional empirical data emerge. Third, behavioural change was modelled as a uniform reduction in sexual activity during symptomatic periods, whereas in reality, behaviour can be heterogeneous, dynamic, and difficult to sustain. Symptom-triggered measure may be limited by atypical or non-specific symptom recognition, and varying health literacy (May et al., 2023). In addition, prior studies suggest that vaccination may be associated with risk compensation, whereby some individuals feel more comfortable engaging in sexual activity, and abstinence-only approach is often ineffective and hard to maintain over time (Phillips et al., 2024). Thus, our results can inform recommendations on behavioural risk reduction, although the achievable magnitude and duration of change are uncertain and likely context-dependent. In real-world application, more feasible risk-reduction likely depends on supportive, non-stigmatizing health promotion that emphasizes practical harm-reduction messages (e.g., symptom awareness, timely testing and care, and positive risk communication), delivered through timely, accurate information from trusted official sources, rather than relying solely on sexual abstinence (May et al., 2023; Phillips et al., 2024). Fourth, we did not model waning immunity (Faherty et al., 2024), which could influence longer-term dynamics. Finally, this study focused on Thailand as a case study; while the findings are broadly relevant, transmission dynamics and intervention effectiveness may differ in other contexts. Future research should address these limitations by incorporating more detailed behavioural heterogeneity, exploring additional routes of transmission, and validating model outcomes with real-world epidemiological and vaccination data as they become available. Evaluating the cost-effectiveness of alternative strategies, particularly combined vaccination and behavioural interventions, will also be important for guiding public health decision-making.

In conclusion, our study suggest that under constrained vaccine supply, PrEP could provide the greatest potential benefit when implemented early, while PEP could be highly effective when aligned with periods of high incidence. Mixed strategies and dose-sparing approaches can extend the reach of limited vaccines, and even modest behavioural changes may substantially enhance epidemic control. These findings reinforce the need for flexible, context-specific intervention packages that integrate vaccination and behavioural measures to reduce Mpox transmission among GBMSM in Thailand and similar settings.

CRedit authorship contribution statement

Sutham Jirapanakorn: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bannaporn Hunsin:** Writing – review & editing, Visualization, Validation, Software, Formal analysis, Data curation. **Akira Endo:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Clement Lee:** Writing – review & editing, Validation, Investigation, Data curation. **Shihui Jin:** Writing – review & editing, Validation. **Borame Lee Dickens:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Thitiya Theparod:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Ethical approval

No specific ethics approval was required for this article.

Data availability statement

All data used in this study are publicly available from the Thailand Department of Disease Control (<https://ddc.moph.go.th/monkeypox/dashboard.php>).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research project was financially supported by Mahasarakham University. A.E. reports support from the Japan Science and Technology Agency (JPMJPR22R3), Japan Agency for Medical Research and Development (JP223fa627004), and the Japan Society for the Promotion of Science (JP22K17329).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2026.03.002>.

References

- Agency, U. H. S. (2025). Mpox: Technical assessments [Internet]. U. Government <https://assets.publishing.service.gov.uk/media/6901ee58918e1f940b3cf79c/mpox-technical-assessment-27-october-2025.pdf>.
- Alakunle, E., Moens, U., Nchinda, G., & Okeke, M. I. (2020). Monkeypox virus in Nigeria: Infection biology, epidemiology, and evolution. *Viruses*, 12(11), 1257. <https://doi.org/10.3390/v12111257>
- BBC News. (2024). Thailand confirms first Asian case of new mpox strain. August 22, 2024, Retrieved 4 June from <https://www.bbc.com/news/articles/czrgpg127zgo>.
- Brooks, J. T., Marks, P., Goldstein, R. H., & Walensky, R. P. (2022). Intradermal vaccination for monkeypox — Benefits for individual and public health. *New England Journal of Medicine*, 387(13), 1151–1153. <https://doi.org/10.1056/NEJMp2211311e>
- Centers for Disease Control and Prevention. (2024a). *Signs and symptoms of mpox*, 12 September 2024. Atlanta: CDC. Retrieved 25 Nov from <https://www.cdc.gov/mpox/signs-symptoms/index.html>.
- Centers for Disease Control and Prevention. (2024b). *About mpox*, 29 October 2024 <https://restoredcdc.org/www.cdc.gov/mpox/about/index.html>.
- Centers for Disease Control and Prevention. (2025). What to Do if you are sick, 12 December 2025 <https://www.cdc.gov/monkeypox/caring/index.html>.
- Dalton, A. F. D. A., Chard, A. N., Moulia, D. L., Deputy, N. P., Fothergill, A., Kracalik, I., Wegner, C. W., Markus, T. M., Pathela, P., Still, W. L., Hawkins, S., Mangla, A. T., Ravi, N., Licherdel, E., Britton, A., Lynfield, R., Sutton, M., Hansen, A. P., & Wyatt, D. (2023). Estimated effectiveness of JYNNEOS vaccine in preventing Mpox: A multijurisdictional case-control study — United States, August 19, 2022–March 31, 2023. *MMWR. Morbidity and Mortality Weekly Report*, 72(20), 553–558. <https://doi.org/10.15585/mmwr.mm7220a3>
- Department of disease control. (2025). *Mpox situation report in Thailand*, 3 May 2025. Thailand: Department of Disease Control, Ministry of Public Health. Retrieved 10 Feb from <https://ddc.moph.go.th/monkeypox/dashboard.php>.
- Deputy, N. P., Deckert, J., Chard, A. N., Sandberg, N., Moulia, D. L., Barkley, E., Dalton, A. F., Sweet, C., Cohn, A. C., Little, D. R., Cohen, A. L., Sandmann, D., Payne, D. C., Gerhart, J. L., & Feldstein, L. R. (2023). Vaccine effectiveness of JYNNEOS against mpox disease in the United States. *New England Journal of Medicine*, 388(26), 2434–2443. <https://doi.org/10.1056/NEJMoa2215201>
- Dimitrov, D., Adamson, B., & Matrajt, L. (2023). Evaluation of mpox vaccine dose-sparing strategies. *PNAS Nexus*, 2(5). <https://doi.org/10.1093/pnasnexus/pgad095>
- Faherty, E. A. G., Holly, T., Ogale, Y. P., Spencer, H., Becht, A. M., Crisler, G., Wasz, M., Stonehouse, P., Barbian, H. J., Zelinski, C., Kittner, A., Foulkes, D., Anderson, K. W., Evans, T., Nicolae, L., Staton, A., Hardnett, C., Townsend, M. B., Carson, W. C., ... Tabidze, I. (2024). Investigation of an mpox outbreak affecting many vaccinated persons in Chicago, Illinois—march 2023–June 2023. *Clinical Infectious Diseases*, 79(1), 122–129. <https://doi.org/10.1093/cid/ciae181>
- Frey, S. E., Wald, A., Edupuganti, S., Jackson, L. A., Stapleton, J. T., Sahly, H. E., El-Kamary, S. S., Edwards, K., Keyserling, H., Winokur, P., Keitel, W., Hill, H., Goll, J. B., Anderson, E. L., Graham, I. L., Johnston, C., Mulligan, M., Rouphael, N., Atmar, R., ... Belshe, R. B. (2015). Comparison of lyophilized versus liquid Modified Vaccinia Ankara (MVA) formulations and subcutaneous versus intradermal routes of administration in healthy vaccinia-naïve subjects. *Vaccine*, 33(39), 5225–5234. <https://doi.org/10.1016/j.vaccine.2015.06.075>
- Gostin, L. O., Jha, A. K., & Finch, A. (2024). The mpox global health emergency — A time for solidarity and equity. *New England Journal of Medicine*, 391(14), 1265–1267. <https://doi.org/10.1056/NEJMp2410395>
- Government of Canada. (2025). Mpox: Symptoms, testing, care and treatment, 13 March 2025 <https://www.canada.ca/en/public-health/services/diseases/mpox/symptoms-management.html>.
- Government of Western Australia Department of Health. (2025). Mpox (monkeypox) virus infection. Retrieved 15 Dec from https://www.health.wa.gov.au/Articles/J_M/Monkeypox-virus-infection.
- Granskog, L., Saadeh, K., Lorenz, K., Quint, J., Salih, T., Lo, T., Jacobson, K., Ramos, M., Chapman, E., Snyder, R. E., & Lewnard, J. A. (2025). Effect of JYNNEOS vaccination on mpox clinical progression: A case-control study. *The Lancet Infectious Diseases*, 25(5), 456–465. [https://doi.org/10.1016/S1473-3099\(25\)00180-X](https://doi.org/10.1016/S1473-3099(25)00180-X)
- Huang, Q., Sun, Y., Jia, M., Jiang, M., Xu, Y., Feng, L., & Yang, W. (2024). An effectiveness study of vaccination and quarantine combination strategies for containing mpox transmission on simulated college campuses. *Infectious Disease Modelling*, 9(3), 805–815. <https://doi.org/10.1016/j.idm.2024.04.004>
- Jirapanakorn, S., Sornsirin, W., Sangthong, P., Pisitpayat, N., & Suphanchaimat, R. (2024). An investigation of the first non-sexual household-transmitted mpox cluster with rapid situation analysis of mpox epidemic in Thailand. *Outbreak, Surveillance, Investigation & Response (OSIR) Journal*, 17(1), 36–44. <https://doi.org/10.59096/osir.v17i1.267118>

- Kottkamp, A. C., Samanovic, M. I., Duerr, R., Oom, A. L., Belli, H. M., Zucker, J. R., Rosen, J. B., & Mulligan, M. J. (2023). Antibody titers against mpox virus after vaccination. *New England Journal of Medicine*, 389(24), 2299–2301. <https://doi.org/10.1056/NEJMc2306239>
- Kröger, S. T., Lehmann, M. C., Treutlein, M., Fiethe, A., Kossow, A., Küfer-Weiß, A., Nießen, J., & Grüne, B. (2023). Mpox outbreak 2022: An overview of all cases reported to the cologne health department. *Infection*, 51(5), 1369–1381. <https://doi.org/10.1007/s15010-023-01997-x>
- Lee, Y.-L., Chen, G.-J., Chen, N.-Y., Liou, B.-H., Wang, N.-C., Lee, Y.-T., Yang, C.-J., Huang, Y.-S., Tang, H.-J., Huang, S.-S., Lin, Y.-C., Cheng, C.-Y., Lee, C.-H., Chen, T.-C., Wu, T.-S., Liu, C.-E., Lu, P.-L., Hung, C.-C., & Group, T. H. S. (2018). Less severe but prolonged course of acute hepatitis A in Human Immunodeficiency Virus (HIV)-infected patients compared with HIV-uninfected patients during an outbreak: A multicenter observational study. *Clinical Infectious Diseases*, 67(10), 1595–1602. <https://doi.org/10.1093/cid/ciy328>
- Lin, Y.-C., Wen, T.-H., Shih, W.-L., Vermund, S. H., & Fang, C.-T. (2024). Impact of vaccination and high-risk group awareness on the mpox epidemic in the United States, 2022–2023: A modelling study. *eClinicalMedicine*, 68, Article 102407. <https://doi.org/10.1016/j.eclinm.2023.102407>
- macroTrends. (n.d.). *Thailand life expectancy 1950–2025*. macroTrends. Retrieved 23 Jan from <https://www.macroTrends.net/global-metrics/countries/thailand/life-expectancy>
- Maniscalco, D., Robineau, O., Boëlle, P.-Y., Mazzoli, M., Barret, A.-S., Chazelle, E., Mailles, A., Noël, H., Tarantola, A., Velter, A., Zanetti, L., & Colizza, V. (2024). Adaptive behavior in response to the 2022 mpox epidemic in the Paris region. *medRxiv*. <https://doi.org/10.1101/2024.10.25.24315987>
- Mason, L. M. K., Betancur, E., Riera-Montes, M., Lienert, F., & Scheele, S. (2024). MVA-BN vaccine effectiveness: A systematic review of real-world evidence in outbreak settings. *Vaccine*, 42(26), Article 126409. <https://doi.org/10.1016/j.vaccine.2024.126409>
- May, T., Towler, L., Smith, L. E., Horwood, J., Denford, S., Rubin, G. J., Hickman, M., Amlôt, R., Oliver, I., & Yardley, L. (2023). Mpox knowledge, behaviours and barriers to public health measures among gay, bisexual and other men who have sex with men in the UK: A qualitative study to inform public health guidance and messaging. *BMC Public Health*, 23(1), 2265. <https://doi.org/10.1186/s12889-023-17196-0>
- Mrsny, R. J. (2022). Does an intradermal vaccination for monkeypox make sense? *The AAPS Journal*, 24(6), 104. <https://doi.org/10.1208/s12248-022-00754-6>
- Okoli, G. N., Van Caesele, P., Askin, N., & Abou-Setta, A. M. (2023). Comparative evaluation of the clinical presentation and epidemiology of the 2022 and previous Mpox outbreaks: A rapid review and meta-analysis. *Infectious Diseases*, 55(7), 490–508. <https://doi.org/10.1080/23744235.2023.2214609>
- Okoli, G. N., Van Caesele, P., Askin, N., & Abou-Setta, A. M. (2024). A global systematic evidence review with meta-analysis of the epidemiological characteristics of the 2022 Mpox outbreaks. *Infection*, 52(3), 901–921. <https://doi.org/10.1007/s15010-023-02133-5>
- Oliveira, G. P., Rodrigues, R. A. L., Lima, M. T., Drummond, B. P., & Abrahão, J. S. (2017). Poxvirus host range genes and virus–host spectrum: A critical review. *Viruses*, 9(11), 331. <https://doi.org/10.3390/v9110331>
- Pang, Y., Cao, D., Zhu, X., Long, Q., Tian, F., Long, X., & Li, Y. (2024). Safety and efficacy of the modified vaccinia Ankara–Bavaria nordic vaccine against mpox in the real world: Systematic review and meta-analysis. *Viral Immunology*, 37(4), 216–219. <https://doi.org/10.1089/vim.2023.0147>
- Phillips, G., 2nd, Curtis, M. G., Felt, D., Davoudpour, S., Rodriguez-Ortiz, A. E., Cortez, A., French, A. L., Hosek, S. G., & Serrano, P. A. (2024). Changes in sexual behaviors due to mpox: A cross-sectional study of sexual and gender minority individuals in Illinois. *Prevention Science: The Official Journal of the Society for Prevention Research*, 25(4), 628–637. <https://doi.org/10.1007/s11121-023-01604-3>
- Reda, A., El-Qushayri, A. E., & Shah, J. (2023). Asymptomatic monkeypox infection: A call for greater control of infection and transmission. *The Lancet Microbe*, 4(1), e15–e16. [https://doi.org/10.1016/s2666-5247\(22\)00259-2](https://doi.org/10.1016/s2666-5247(22)00259-2)
- Richards, J., Krantz, E., Selke, S., & Wald, A. (2008). Healthcare seeking and sexual behavior among patients with symptomatic newly acquired genital herpes. *Sexually Transmitted Diseases*, 35(12), 1015–1021. <https://doi.org/10.1097/OLQ.0b013e318182a596>
- Shen, Y.-H., Chang, H.-H., Tou, C.-Y., Lee, Y.-C., Lee, Y.-H., Cheng, S.-Y., Huang, K.-C., & Lu, C.-W. (2024). Stigmatization and preferences in monkeypox vaccine regimens. *Archives of Sexual Behavior*, 53(10), 3825–3834. <https://doi.org/10.1007/s10508-024-02975-6>
- Srivastava, S., Laxmi, Sharma, K., Sridhar, S. B., Talath, S., Shareef, J., Mehta, R., Satapathy, P., & Sah, R. (2024). Clade Ib: A new emerging threat in the Mpox outbreak. *Frontiers in Pharmacology*, 15, Article 1504154. <https://doi.org/10.3389/fphar.2024.1504154> [Review].
- Titanji, B. K., Tegomoh, B., Nematollahi, S., Konomos, M., & Kulkarni, P. A. (2022). Monkeypox: A contemporary review for healthcare professionals. *Open Forum Infectious Diseases*, 9(7). <https://doi.org/10.1093/ofid/ofac310>
- Tiwari, A., Kalonji, T., Miller, T., Van Den Bossche, T., Krollicka, A., Muhindo-Mavoko, H., Mitashi, P., Tahita, M. C., Lood, R., Pitkänen, T., & Maketa, V. (2025). Emergence and global spread of mpox clade Ib: Challenges and the role of wastewater and environmental surveillance. *The Journal of Infectious Diseases*, 231(5), e825–e829. <https://doi.org/10.1093/infdis/jiaf006>
- Wenham, C., & Eccleston-Turner, M. (2022). Monkeypox as a PHEIC: Implications for global health governance. *The Lancet*, 400(10369), 2169–2171. [https://doi.org/10.1016/S0140-6736\(22\)01437-4](https://doi.org/10.1016/S0140-6736(22)01437-4)
- Wipatayotin, A. (2024). B21 million approved for mpox vaccines. *Bangkok Post*. Retrieved 25 Nov from <https://www.bangkokpost.com/thailand/general/2860827/b21-million-approved-for-mpox-vaccines>
- World Health Organization. (2022). *Vaccines and immunization for monkeypox: Interim guidance, 16 November 2022*. World Health Organization. Retrieved 25 Nov from <https://www.who.int/publications/i/item/WHO-MPX-Immunization>
- World Health Organization. (2024). *Mpox*, 26 August 2024. World Health Organization. Retrieved 25 Nov from <https://www.who.int/news-room/fact-sheets/detail/mpox>